

# Jamal Ghobrial

AI Engineer | Machine Learning | LLM Agents & RAG Systems

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## Professional Summary

AI Engineer with a BSc in Bioinformatics (GPA 3.61/4.0) and hands-on experience across 3 internships delivering production-grade LLM agents, RAG pipelines, multimodal search platforms, and Learning-to-Rank systems. Skilled in the full AI lifecycle — data engineering, model development, and cloud deployment (Azure, Docker, Modal). Top 29% Kaggle competitor; certified by NVIDIA, Microsoft, and WorldQuant University.

## Technical Skills

**Languages & Libraries:** Python (PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, SciPy), SQL, Go, C++

**AI / LLMs / RAG:** RAG Pipelines, LangGraph, Agentic Workflows, Prompt Engineering, ChromaDB, Qdrant, LibreChat, MoA

**Search & IR:** Learning-to-Rank (LTR), LightGBM LambdaRank, BM25 / Tantivy, Hybrid Search (RRF), SentenceTransformers

**CV & Speech:** YOLO11 Pose Estimation, DINO Visual Embeddings, Speech-to-Text (Groq / Whisper), Image Processing

**Frameworks & Web:** Hugging Face Transformers, FastAPI, Express.js, React, Node.js, Streamlit

**Cloud & MLOps:** Microsoft Azure, AWS SageMaker, Docker, Modal (Serverless GPU), Git/GitHub

## Experience

**AI Engineer Intern** — Areeb Technology, Cairo EG

- Built **OmniFind**, a full-stack multimodal e-commerce search platform (text, DINO+Qdrant visual, Whisper audio) indexing 9,800+ products; containerised with Docker + MongoDB.
- Developed **Semantrix**, a 2-stage LTR search engine: Tantivy BM25 recall re-ranked with LightGBM LambdaRank (7 engineered features) deployed serverlessly via Modal.
- Architected **ContentPro Creator Agent**: a 3-model MoA pipeline (Gemini research → Claude Sonnet writing → GPT-4.1 editing) on LibreChat with Tavily web search and RAG.
- Streamlined Azure-hosted team workflows using Git/GitHub structured PRs and code reviews.

**AI Engineer Intern** — NTI, Cairo EG

- Designed **Digitopia AI Competition Judgment System**: 2-phase pipeline using Groq STT + YOLO11 pose estimation (Phase 1) and LangGraph Plan→Score→Verify grading agent with PDF/PPTX parsing (Phase 2).
- Built a ChromaDB-backed RAG assistant for natural-language querying of results plus automated PDF report export.
- Benchmarked ML, Deep Learning, CV, and NLP models across PyTorch and TensorFlow with structured performance reports.

**ML Engineer Intern (Microsoft Track)** — DEPI, Cairo EG

- Developed classification, regression, and clustering models with Scikit-Learn and TensorFlow; applied MLOps best practices for data preprocessing, validation, and model tracking.
- Automated ETL pipelines to reduce training data preparation time and improve reproducibility.

**Machine Learning Intern** — ITIDA / EME Initiative, Assiut EG

- Built an industrial defect prediction model (XGBoost + Scikit-Learn) achieving **85% accuracy** with a **40% performance uplift** via dimensionality reduction and hyperparameter tuning.
- Placed **Top 29%** and **Top 33%** globally in competitive Kaggle ML challenges.

## Featured Projects

**Digitopia AI Competition Judgment System** 🌐

- Automated hackathon judging via multimodal analysis (audio STT, body-language pose, PDF/PPTX) and a LangGraph agentic grader with ChromaDB RAG and leaderboard dashboard.

**LexAI: Legal Document Assistant** 🌐

- Zero-wrapper legal RAG pipeline with Hybrid Retrieval (RRF), section-aware chunking (Articles/Clauses), SSE streaming, source citations, cross-document comparison, and DOCX export.

**Semantrix: Learning-to-Rank Search Engine**

- 2-stage LTR engine on MS MARCO: Tantivy BM25 recall re-ranked with LightGBM LambdaRank (7 engineered features) and Modal serverless GPU inference.

**OmniFind: Multi-Modal Product Discovery Platform**

- Full-stack e-commerce search platform supporting text, DINO+Qdrant visual similarity, and Whisper audio search indexing 9,800+ products.

**ContentPro Creator Agent**

- 3-model Mixture-of-Agents (MoA) content pipeline on LibreChat: Gemini web search research → Claude Sonnet writing → GPT-4.1 formatting and quality control.

**Education**

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**BSc in Bioinformatics** — GPA **3.61 / 4.0** (Very Good with Honors)  
*Faculty of Computers & Information, Assiut University* — Graduation Project recognised for excellence.

**Courses & Certifications**

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|---------------------------------------------------------|----------|
| <b>Getting Started with Deep Learning</b> — NVIDIA DLI  | Dec 2025 |
| <b>Applied Data Science Lab</b> — WorldQuant University | Jul 2023 |
| <b>Data Analysis Professional Nanodegree</b> — Udacity  | Dec 2022 |

**Additional Information**

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**Languages:** Arabic (Native)    English (Professional Working)    **Military Service:** Completed (Jun 2025)